

Scalability & Future Plan

Date	3 JULY 2024
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to local deployment	Cannot be accessed remotely by many users	High
2	No user authentication	User data and history are not secured	High
3	Basic storage using local files	Difficult to manage large amounts of data	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users	Deploy on cloud with load balancing
2	Data Storage	Local JSON/text storage	Use MySQL or PostgreSQL database
3	Performance	Single-server processing	Optimize APIs and use caching
4	Security	Basic application security	Add user authentication, HTTPS, and JWT security

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Login & Authentication	1 Month	Secure user access
Phase 3	Cloud Deployment & Database Integration	2 Months	Better scalability and reliability
Phase 4	AI Chatbot with Personalized Learning Recommendations	3 Months	Improved learning experience and intelligent assistance